


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
16-05-2023
JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

SPACE FOR ROUGH WORK

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

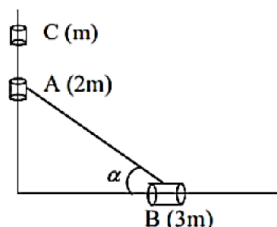
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- Two collars, A of mass 2m and B of mass 3m are connected by a massless rod with pinjoints so that they are constrained to move along the fixed perpendicular rods on horizontal plane as shown, in the accompanying figure. Another collar C of mass m hits A with speed 'v' and sticks to it. There is no friction between collars and rods.



- The speed of collar C just after the impact is $\frac{v \cos^2 \alpha}{3}$
- The speed of collar A just after the impact is $\frac{v \sin^2 \alpha}{3}$
- The speed of collar B just after the impact is $\frac{v \sin 2\alpha}{6}$
- The speed of collar B just after the impact is $\frac{v \cos 2\alpha}{3}$

2. A non-relativistic particle of mass m is held in a circular orbit around the origin by an attractive force. The potential energy of the force is given by $U(r) = \frac{kr^2}{2}$, where r is the distance of the particle from origin and k is a constant. Assuming the Bohr quantization of the angular momentum of the particle, choose the **correct** ($v \rightarrow$ velocity of particle, $KE(r) \rightarrow$ kinetic energy of particle, $E(r) \rightarrow$ total energy of the particle)

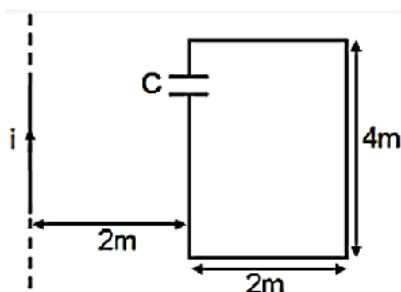
(A) $r^2 = \frac{nh}{2\pi\sqrt{km}}$

(B) $v^2 = \frac{nh}{2\pi\sqrt{km}}$

(C) $KE(r) = U(r)$

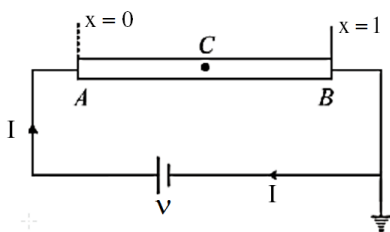
(D) $E(r) = \frac{nh}{2\pi} \sqrt{\frac{k}{m}}$

3. A rectangular frame of negligible resistance with capacitor of capacitance C is placed nearby an infinitely long wire carrying current $I = I_0 \sin\left(\omega t + \frac{\pi}{6}\right)$, wire and frame being in same plane. Select correct alternative(s):



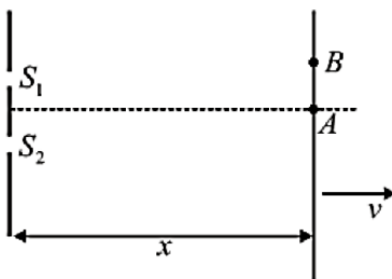
- (A) Maximum charge on capacitor is $\left(\frac{4\mu_0 i_0 \omega \ln 2}{\pi}\right) C$
- (B) Charge on capacitor at $t = 0$ is $\left(\frac{\sqrt{3}\mu_0 i_0 \omega \ln 2}{\pi}\right) C$
- (C) Current in frame is minimum at $t = \frac{\pi}{3\omega}$
- (D) Maximum current in frame is $\left(\frac{2\omega^2 \mu_0 i_0 \ln 2}{\pi}\right) C$

4. In a resistance wire of unit length and unit cross sectional area, resistivity varies as $P = \frac{P_0}{1+x}$. This wire is connected across a battery of emf V .



A and B are two ends of the wire and C is mid- point. Choose the **correct** options ?

- (A) current $I = \frac{V}{P_0 \ln 2}$ (B) Electric field at A = $\frac{V}{\ln 2}$
 (C) Electric field at B = $\frac{3V}{2 \ln 2}$ (D) Electric field at C = $\frac{2V}{3 \ln 2}$
5. In Young's double slit experiment at time $t = 0$, A is central maxima and B is 4th maxima. Separation between slits and screen is increasing at a constant speed v . Then for a later time ' t '. Choose the **correct** option.



- (A) At $t = \frac{x}{v}$, B will be 2nd order maxima (B) At $t = \frac{2x}{v}$, B will be 1st maxima
 (C) At $t = \frac{5x}{3v}$, B will be 2nd minima (D) At $t = \frac{7x}{v}$, B will be 1st minima

6. A space capsule is in a circular orbit of radius r_C about the earth. A short burst from a rocket in the backward direction increases the kinetic energy by 20% and causes the capsule to enter an elliptical orbit with the point of firing as the perigee; i.e., $r_p = r_C$.

Then choose the **CORRECT** statement(s) for further motion of capsule:- (Assume mass of capsule to remain nearly constant)

- (A) Total mechanical energy of capsule decreases.
- (B) Time period of capsule will decreases.
- (C) Distance of apogee from centre of earth is $r_A = 1.5r$
- (D) If mechanical energy before the event (short burst) is E_1 & final mechanical energy after the event is E_f , then $E_f = 0.8E_i$.

SECTION-I (ii) : (Maximum Marks: 12)

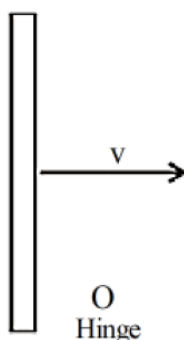
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. A uniform rod of length L is moving with speed v on a horizontal smooth surface. One end of the rod is suddenly hinged - this process takes a very small time $L/100v$. Match the ratios in list I with their values in list II.

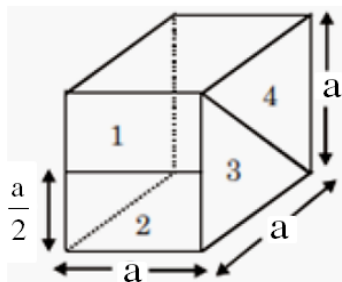


	List-I		List-II
(P)	Speed of the centre of mass of the rod just after to just before the event.	(1)	2/3
(Q)	KE of the rod just after to just before	(2)	2/9
(R)	Average force exerted by the hinge during hinging to (100x force exerted by hinge just after hinging)	(3)	3/4
(S)	Time taken to turn the rod by one radian after the event to time taken to cover the length of the rod before the event	(4)	5/6

Code:

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$
 (B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$
 (C) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$
 (D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

8. A liquid of density ρ is completely filled in a cubical container as shown in figure. Neglect the atmospheric pressure, match the quantities in List-I with corresponding expression in List-II and select the **CORRECT** code



	List-I		List-II
(P)	Force on rectangular Face 1 is	(1)	$\frac{1}{3}a^3\rho g$
(Q)	Force on rectangular Face 2 is	(2)	$\frac{1}{8}a^3\rho g$
(R)	Force on triangular Face 3 is	(3)	$\frac{1}{6}a^3\rho g$
(S)	Force on triangular Face 4 is	(4)	$\frac{3}{8}a^3\rho g$

Code:

- (A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$
 (B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$
 (C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$
 (D) $P \rightarrow 2, 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

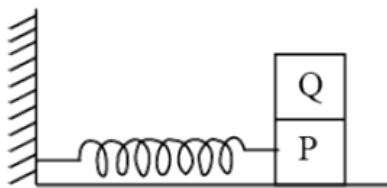
9. The electric field E is measured at a point $P(0,0,d)$ generated due to various charge distributions and the dependence of E on d is found to be different for different charge distributions. List-I contains different relations between E and d . List-II describes different electric charge distributions, along with their locations. Match the functions in List-I with the related charge distributions in List-II.

	List-I		List-II
(P)	E is independent of d	(1)	A point charge Q at the origin
(Q)	$E \propto \frac{1}{d}$	(2)	A small dipole with point charges Q at $(0,0,\ell)$ and $-Q$ at $(0,0,-\ell)$. Take $2\ell \ll d$
(R)	$E \propto \frac{1}{d^2}$	(3)	An infinite line charge coincident with the x -axis, with uniform linear charge density λ .
(S)	$E \propto \frac{1}{d^3}$	(4)	Two infinite wires carrying uniform linear Charge density parallel to the x -axis. The One along $(y = 0, z = \ell)$ has a charge density $+\lambda$ and the one along $(y = 0, z = -\ell)$ has a charge density $-\lambda$. Take $2\ell \ll d$
		(5)	Infinite plane charge coincident with the xy -plane with uniform surface charge density

Code:

- (A) $P \rightarrow 5; Q \rightarrow 3,4; R \rightarrow 1; S \rightarrow 2$
 (B) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 1,4; S \rightarrow 2$
 (C) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 1,2; S \rightarrow 4$
 (D) $P \rightarrow 4; Q \rightarrow 2,3; R \rightarrow 1; S \rightarrow 5$

10. A block P of mass m is placed on smooth horizontal surface and connected with a spring. Another block Q of same mass is placed on the block P. The two blocks are pulled by a distance A and released. The blocks oscillates without slipping on each other. Then match the matrix.



	List-I		List-II
(P)	Friction force on the block Q, when both the blocks are coming towards equilibrium position from left	(1)	Towards right
(Q)	Friction force on the block Q when both block going right from equilibrium position (exclude the equilibrium position)	(2)	Towards left
(R)	Friction force will be maximum on Q	(3)	At equilibrium position
(S)	Friction force on P when both blocks are coming towards the equilibrium position from right	(4)	At extreme position
		(5)	None of these

Code:

- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$
 (B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$
 (C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$
 (D) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

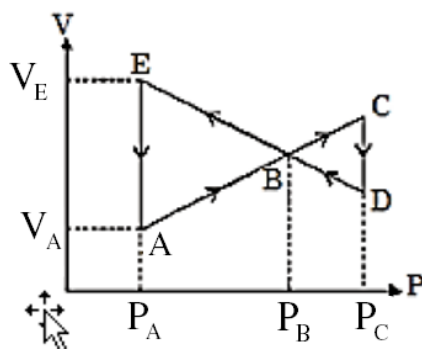
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

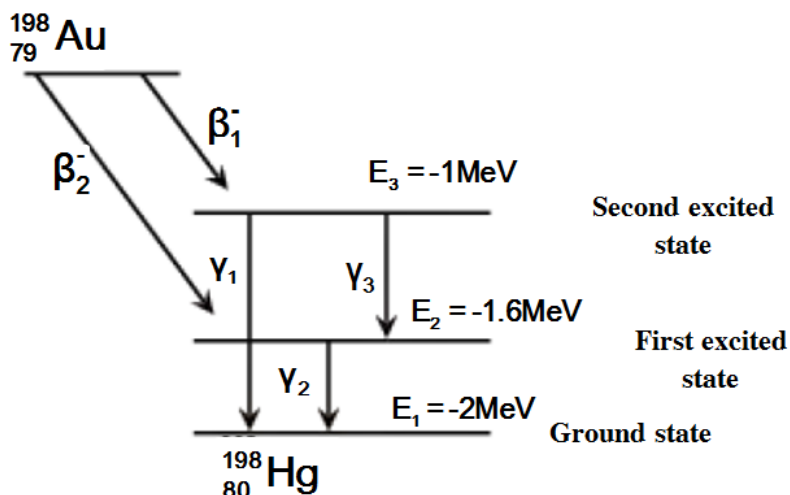
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

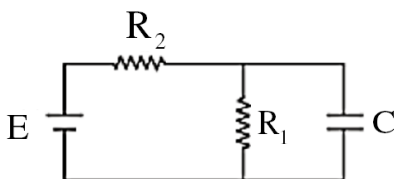
- Figure shows a cyclic process ABCDBEA performed on an ideal gas. If $P_A = 2 \text{ atm.}$, $P_B = 4.5 \text{ atm.}$ and $P_C = 6 \text{ atm.}$; $V_E - V_A = 0.05 \text{ litre}$, find the magnitude of work done by the gas (in J) in the complete process in nearest integer.
(1 atm, pressure = $1 \times 10^5 \text{ pa}$)



2. Gold Nucleus(${}_{79}\text{Au}^{198}$) can decay into mercury nucleus (${}_{80}\text{Hg}^{198}$) by two decay schemes shown in figure, (i) It can emit a β particle (β_1) and come to ground state by either emitting one γ ray (γ_1) or emitting two γ rays (γ_3 & γ_2). (ii) It can emit one β particle (β_2) and come to ground state by emitting γ_2 ray. Atomic masses: ${}^{198}\text{Au} = 197.9682$ amu, ${}^{198}\text{Hg} = 197.9662$ amu, $1 \text{ amu} = 930 \text{ MeV}/c^2$. The energy levels of the nucleus are shown in figure. The maximum kinetic energy of emitted β_2 particles in MeV is equal to



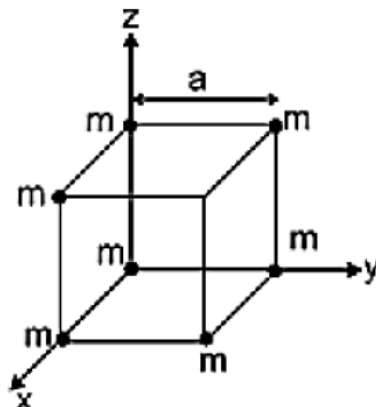
3. For the given circuit as shown in figure.



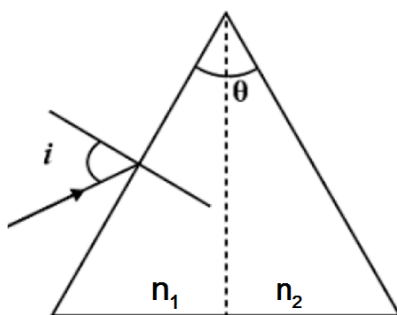
$$R_1 = (60 \pm 3) \Omega \quad R_2 = (30 \pm 6) \Omega \quad C = (1 \pm 0.01) \mu\text{F} \quad E = (40 \pm 4) \text{ V}$$

At $t = 0$ capacitor is uncharged. Maximum fractional error in maximum charge stored in capacitor is _____ .

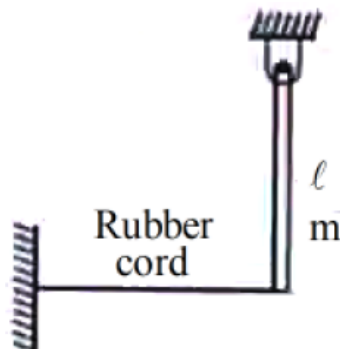
4. Seven particles, each of mass m are placed at the seven corners of a cube of side ' a ', but one corner is vacant, as shown in figure. If the x-coordinate of the centre of mass of the system is $\frac{na}{m}$. Where n and m are least integer values. Then find value of $(m - n)$.



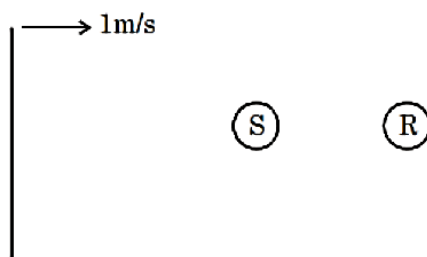
5. For a prism of prism angle $\theta = 60^\circ$, the refractive indices of the left half and the right half are, respectively, n_1 and n_2 ($n_2 \geq n_1$) as shown in the figure. The angle of incidence ' i ' is chosen such that the incident light rays will have minimum deviation if $n_1 = n_2 = n = 1.5$. For the case of unequal refractive indices, $n_1 = n$ and $n_2 = n + \Delta n$ (where $\Delta n < n$), the angle of emergence $e = i + \Delta e$. If $\Delta n = 2.8 \times 10^{-3}$ rad then the value of Δe in milliradians is equal to (Round off to nearest decimal up to 2 digits)



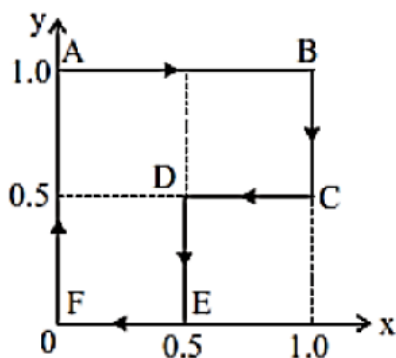
6. The figure shows an oscillatory system (A uniform rod with an elastic string). The system oscillates in vertical plane shown here. The rubber cord is in natural length when rod is vertical. What is the time period of small oscillation (in sec) (k = force constant of rubber cord). Take $k\ell = mg$, $g = \pi^2$, $\ell = 6\text{m}$.



7. A point source and a receiver are stationary. Source sends a wave of frequency 256 Hz, which reaches to receiver directly and also after being reflected from a wall. The wall is moving at speed of 1 m/s. Find beat frequency (in Hz) heard by receiver. Velocity of sound = 321 m/s



8. A particle is moved along a path AB-BC-CD-DE-EF-FA, as shown in figure, in presence of a force $\vec{F} = (\alpha y\hat{i} + 2\alpha x\hat{j})$ N, where x and y are in meter and $\alpha = -1 \text{ N/m}^{-1}$. The work done on the particle by this force $\vec{F}$ will be _____ Joule.



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

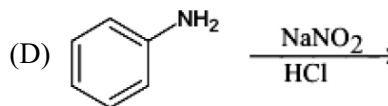
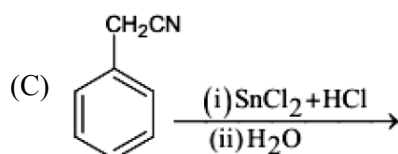
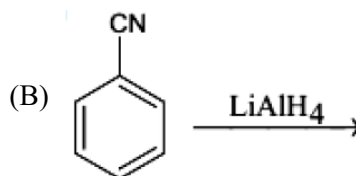
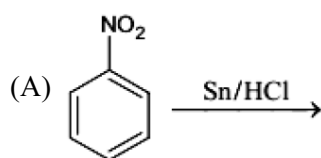
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following is/are **correct**?

- (A) Leaching is the method to remove quantitatively Fe_2O_3 from red bauxite.
- (B) In the metallurgy of Aluminium, purified Al_2O_3 is mixed with Na_3AlF_6 or CaF_2 which lowers the melting point of the mixture and also increases conductivity.
- (C) Chalcocite and cinnabar are sulphide minerals
- (D) Self reduction is not possible during metallurgy of Mg.

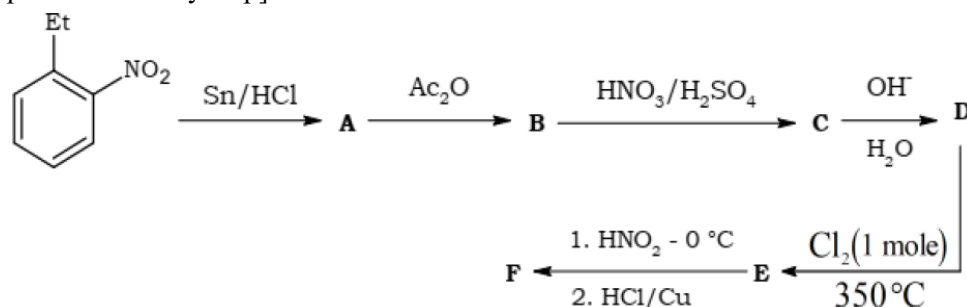
2. Which of the following major organic products produced in the given reactions fails to respond positively in Kjeldahl method of Nitrogen estimation?



3. The **CORRECT** statement(s) about the Group-I metals is/are

- (A) Reactivity of Group-I metals towards water decreases down the group
(B) Among the Group-I metals, Li spontaneously reacts with N_2 to give a coloured crystalline compound.
(C) All the Group-I metal halides are high-melting colourless crystalline solids
(D) KCl is more soluble in H_2O than NaCl

4. **Correct** statement(s) regarding the following reaction sequence is(are) [Consider only major organic product in every step]



- (A) A on treatment with $CHCl_3/KOH$ gives a foul-smelling compound.
(B) A is basic while B is amphoteric.
(C) C is more reactive than B towards alkaline hydrolysis.
(D) F is less reactive than E towards nucleophilic aromatic substitution.

5. 10 mole of liquid 'A' and 10 mole of liquid 'B' is mixed in a cylindrical vessel containing a piston arrangement. Initially a pressure of 2 atm is maintained on the solution. Now, the piston is raised slowly and isothermally. Assuming A and B to be completely miscible and forming an ideal solution. $P_A^0 = 0.6 \text{ atm}$, $P_B^0 = 0.9 \text{ atm}$, then choose incorrect answers (Given : $\sqrt{2425} = 49.25$, $\sqrt{3} = 1.73$, $\sqrt{2} = 1.41$)
- (A) The pressure below which the evaporation of liquid solution will start is: 1.75 atm
- (B) If $\frac{1}{4}$ th of the total moles of liquid solution taken initially is converted to vapour, then moles of B in the vapour will be : 3.825
- (C) The pressure below which the evaporation of liquid solution will start is : 0.75 atm
- (D) If $\frac{1}{2}$ of the total moles of liquid solution taken initially is converted to vapour, then moles of A in the vapour will be: 2.5
6. For the following reaction $2X + Y \xrightarrow{k} P$. The rate of reaction is $\frac{d[P]}{dt} = k[X]$. Two moles of X are mixed with one mole of Y to make 1.0 L of solution. At 50 s, 0.5 mole of Y is left in the reaction mixture. The correct statement(s) about the reaction is/are (Use: $\ln 2 = 0.693$)
- (A) The rate constant, k, of the reaction is $13.86 \times 10^{-4} \text{ s}^{-1}$
- (B) Half-life of X is 50 s
- (C) At 50s, $-\frac{d[X]}{dt} = 13.86 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$
- (D) At, 100s, $-\frac{d[Y]}{dt} = 3.465 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match the compounds in LIST-I with the observations in LIST-II, and choose the correct option:

	List-I		List-II
(P)	Aniline	(1)	Sodium fusion extract of the compound on boiling with FeSO_4 , followed by acidification with conc. H_2SO_4 , gives Prussian blue colour.
(Q)	o-Cresol	(2)	Sodium fusion extract of the compound on treatment with sodium nitroprusside gives blood red colour.
(R)	Cysteine	(3)	Addition of the compound to a saturated solution of NaHCO_3 results in effervescence.
(S)	Caprolactam	(4)	The compound reacts with bromine water to give a white precipitate.
		(5)	Treating the compound with neutral FeCl_3 solution produces violet colour.

Code:

(A) $P \rightarrow 1,2; Q \rightarrow 4; R \rightarrow 2,3; S \rightarrow 1$

(B) $P \rightarrow 1; Q \rightarrow 3,4; R \rightarrow 3; S \rightarrow 2,4$

(C) $P \rightarrow 2,4; Q \rightarrow 1,5; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 1,4; Q \rightarrow 5; R \rightarrow 2,3; S \rightarrow 1$

8. Match the coloured inorganic compounds in **List-I** with the appropriate phenomenon responsible for the colour in **List-II**:

	List-I		List-II
(P)	PbI ₂ (Yellow)	(1)	Polarisation
(Q)	(NH ₄) ₂ [Ce(NO ₃) ₆] (Orange-red)	(2)	Ligand → Metal charge transfer
(R)	CuSO ₄ ·5H ₂ O (Blue)	(3)	f → f transition
(S)	KMnO ₄ (Purple)	(4)	Metal → Ligand charge transfer
		(5)	d → d transition

Codes:

- (A) P → 1; Q → 2; R → 5; S → 2
 (B) P → 4; Q → 5; R → 4; S → 2
 (C) P → 4; Q → 2; R → 5; S → 4
 (D) P → 1; Q → 4; R → 5; S → 5

9. For the reaction: $A \rightarrow \text{product}$, $t_{x/y}$ represents the times in which x/y fraction of reactant is converted into products.

	List-I		List-II
(P)	$t_{5/9}$	(1)	Equal to 54s, if $t_{1/3}$ is 18s in case of first-order reaction
(Q)	$t_{19/27}$	(2)	Equal to 32s, if $t_{1/4}$ is 16s in case of first-order reaction
(R)	$t_{7/8}$	(3)	Equal to 56s, if $t_{1/3}$ is 4s in case of second-order reaction
(S)	$t_{7/16}$	(4)	Equal to 30s, if $t_{1/3}$ is 18s in case of zero-order reaction.
		(5)	Equal to 28s, if $t_{1/2}$ is 16s in case of zero-order reaction.

Codes:

- (A) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 2$
 (B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 5$
 (C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$
 (D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 5$
10. Match the columns (assume the unit cell edge length equal to a).

	List-I		List-II
(P)	a	(1)	Two times the smallest distance between two similarly charged ions in fluorite structure.
(Q)	$\frac{a}{\sqrt{2}}$	(2)	Distance between the next nearest ions in NaCl structure.
(R)	$\sqrt{3}a$	(3)	Distance between the next nearest ions in CsCl structure.
(S)	$\sqrt{2}a$	(4)	Four times the smallest distance of tetrahedral voids from the corner of FCC unit cell
		(5)	Smallest distance between the nearest similarly charged ions in antiferrofluorite structure.

Codes:

- (A) $P \rightarrow 1,5; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$
 (B) $P \rightarrow 1,5; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$
 (C) $P \rightarrow 2,5; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$
 (D) $P \rightarrow 2,5; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

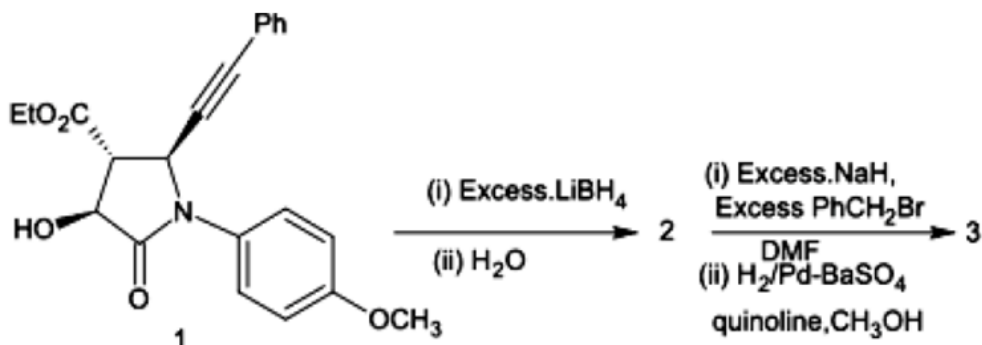
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Lithium borohydride (LiBH_4) is a useful reducing agent, as it is more selective than LiAlH_4 but less selective than NaBH_4 . Unlike the later, LiBH_4 will reduce esters. However, it will not reduce amides. This selectivity is frequency employed by synthetic organic chemists. In the following reaction sequence, 50.6gm of product (3) is obtained with yield of 65% and 75% respectively in 1st and 2nd steps. Find the number of moles of reactant used in this reaction? (C = 12, O = 16, N = 14, H = 1)



2. Consider the following species:
 (i) IF_5 (ii) ICl_4^- (iii) XeO_2F_2
 (iv) NH_2^- (v) BCl_3 (vi) BeCl_2 (monomer)
 (vii) AsCl_4^+ (viii) B(OH)_3 (ix) NO_2^- (x) ClO_2^+

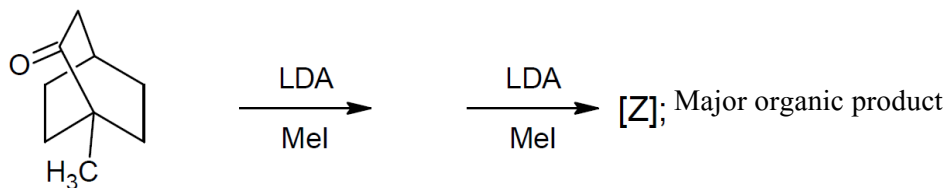
Calculate value of $(X + Y - Z)$, where

X = total number of compounds in which the central atom used their all three p-orbitals in hybridisation respectively.

Y = total number of compounds in which the central atom used only two p-orbitals in hybridisation respectively.

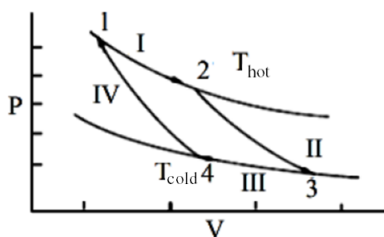
Z = total number of compounds in which the central atom used only one p-orbital in hybridisation respectively.

3. Analyse the following reaction sequence.



If X is the number of alpha hydrogens in the product [Z] and Y is the total no of methyl groups present in the product [Z], then find the value of X + Y.

4. If X is the number of compounds which on strong heating gives oxygen as a product then find the value of X/2.
 HgO , MnO_2 , Pb_3O_4 , KMnO_4 , BaO_2 , LiNO_3 , Ag_2CO_3
5. A pentapeptide has $-\text{COOH}$ group on tyrosine. This produces (Gly), valine (Val), Phenylalanine (Phe), Alanine (Ala) and Tyrosine (Tyr) on complete hydrolysis. For this pentapeptide, the number of possible sequences (primary structures) with $-\text{NH}_2$ group attached to a chiral centre is _____
6. At 143 K. the reaction of XeF_4 with O_2F_2 produces a xenon compound Y. The total number of lone pair(s) of electrons present in the molecule of Y is _____
7. 10 moIs of an ideal gas is participating in the Carnot cycle is shown in the diagram given below.
 Step-I: Isothermal reversible expansion with absorption of q_1 of heat.
 Step-II: Adiabatic reversible expansion.
 Step-III: Isothermal reversible compression with rejection of q_2 of heat.
 Step-IV: Adiabatic reversible expansion.



Calculate the value of $\frac{V_2 V_4}{V_3 V_1}$ in S.I. unit.

8. Hydrogen peroxide solution (20 ml) reacts with a solution of KMnO_4 (20 ml) acidified with dil. H_2SO_4 . The same volume of KMnO_4 solution is just decolorised by 10 ml of MnSO_4 in neutral medium simultaneously forming a dark brown ppt. of hydrated MnO_2 . The Brown ppt. dissolved in 10 ml of 0.2 M sodium oxalate in the presence of dil. H_2SO_4 . If the Molarity of H_2O_2 is P then 10P is

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. If a, b, x, y are positive natural numbers and $\frac{1}{x} + \frac{1}{y} = 1$ then $\frac{a^x}{y} + \frac{b^y}{x}$ is same as :

(A) $\frac{xa^y + yb^x}{x + y}$

(B) $\frac{y^2 \cdot a^x + x^2 \cdot b^y}{x^2 + y^2}$

(C) $\frac{a^x}{x} + \frac{b^y}{y}$

(D) $\frac{a^y}{x} + \frac{b^x}{y}$

2. Let $(1 - x^3)^n = \sum_{r=0}^n a_r x^r (1 - x)^{3n-2r} \forall x \in \mathbb{R} - \{1\}$ then which of the following statement is/are **correct**?
- (A) If $n = 7$ then maximum value of a_r is at $r = 5$ or 6
- (B) $\sum_{r=0}^n a_r$ is divisible by 2^n
- (C) $\frac{a_3}{a_2}$ is integer $\forall n \in \mathbb{N}$
- (D) $\sum_{r=0}^n a_r$ is not divisible by 2^n
3. Let $S(n, k)$ denote the number of ways in which n distinct balls can be put into k identical boxes such that no box remains empty ($k \leq n$). Which of the following is/are correct?
- (A) $S(5, 3) = 25$
- (B) $S(6, 3) = S(5, 3) + S(5, 2)$
- (C) $S(6, 3) = 3S(5, 3) + S(5, 2)$
- (D) The number of onto function from 6 element set of 3 element set is $3! S(6, 3)$

4. If (α, β, γ) is a point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at distance $2\sqrt{7}$ from the point $(1, -1, 0)$, then the coefficient of $x^{\alpha+\beta+\gamma}$ in the expansion of $\left(2x^2 - \frac{1}{x}\right)^9$ is divisible by
- (A) 21 (B) 36
(C) 28 (D) 42
5. Let position vectors of points A, B, C are $\vec{a}, \vec{b}, \vec{c}$ respectively where $\vec{c} = x\vec{a} + y\vec{b}$
- (A) If $x + y = 1, x > 0, y < 0$, then C lies on line AB but not on line segment AB`
(B) If $x > 0, y > 0, x + y < 1$, then C lies inside the triangle $\triangle ABO$ (O is origin)
(C) If $x < 0, y > 0$ and $x + y > 1$, then on plane containing points O, A, B, the points C and origin (O) lies on opposite side of line AB.
(D) If $x \leq 0, y \leq 0$ and $x + y = -1$, then locus of point C is reflection of line segment AB about origin.
6. Let A and B are square matrix of odd order such that $A^T = A^2 - B$ and $B^T = B^2 - A$. Then which of the following statement is/are correct?
- (A) $|B^2 - A^2| = 0$
(B) $|A| \neq |B|$
(C) $|A| = |B|$
(D) $|B^2 - A^2| \neq 0$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match the following:

	List-I		List-II
(P)	The equation of tangent of the ellipse $\frac{x^2}{2} + y^2 = 1$ which cuts off equal lengths of intercepts on coordinate axis is $y = \pm x + a$, then a can be equal to	(1)	$-\sqrt{3}$
(Q)	The normal $y = mx - 2am - am^3$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex then m can be equal to	(2)	$-\sqrt{2}$
(R)	The equation of the common tangent to parabola $y^2 = 4x$ and $x^2 = 4y$ is $x + y + \frac{k}{\sqrt{3}} = 0$, then k is equal to	(3)	$\sqrt{2}$
(S)	An equation of common tangent to parabola $y^2 = 8x$ and the hyperbola $3x^2 - y^2 = 3$ is $2x + \frac{ky}{\sqrt{2}} + 1 = 0$, then k can be equal to	(4)	$\sqrt{3}$

Code:

- (A) $P \rightarrow 1,4; Q \rightarrow 2,3; R \rightarrow 4; S \rightarrow 2,3$
- (B) $P \rightarrow 1,3; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1,4$
- (C) $P \rightarrow 4; Q \rightarrow 1,3; R \rightarrow 2,3; S \rightarrow 1$
- (D) $P \rightarrow 1,4; Q \rightarrow 2,3; R \rightarrow 1; S \rightarrow 3,4$

8. Match the following:

	List-I		List-II
(P)	The number of rational points on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is	(1)	Infinite
(Q)	The number of integral points on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is	(2)	4
(R)	The number of rational points on the ellipse $\frac{x^2}{3} + y^2 = 1$ is	(3)	2
(S)	The number of integral points on the ellipse $\frac{x^2}{3} + y^2 = 1$ is	(4)	0

Code:

- (A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 3$
 (B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$
 (C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 3$
 (D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

9. Match the following:

	List-I		List-II
(P)	Area bounded by $y = x^2 + 2$ and $y = 2 x - \cos \pi x$ is equal to A, then 3A equals	(1)	4
(Q)	The value of $2\cot(\cot^{-1}3 + \cot^{-1}7 + \cot^{-1}13 + \cot^{-1}21)$	(2)	$\frac{8}{3}$
(R)	Tangent and normal at the ends A & C of focal chord AC of parabola $y^2 = 4x$ intersect at B and D, then minimum area of ABCD is	(3)	3
(S)	Number of integral values of ordered pair (α, β) for which the area common to $x^2 + y^2 - 2\alpha^2 x - 2\beta^2 y + c = 0$ and its image in $x + y = 1$ is maximum is	(4)	8

Code:

(A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

10. Let $f(x)$ be a real valued function defined by $f(x) = x^2 - 2|x|$ and

$$g(x) = \begin{cases} \text{minimum } \{f(t) : -2 \leq t \leq x\} & x \in [-2, 0) \\ \text{maximum } \{f(t) : 0 \leq t \leq x\} & x \in [0, 3] \end{cases}$$

	List-I		List-II
(P)	$g(x)$ is not continuous at x equal to	(1)	-2
(Q)	$g(x)$ is not derivable at x equal to	(2)	0
(R)	Number of points of local maxima of $g(x)$ is equal to	(3)	1
(S)	Absolute maximum value of $g(x)$ is equal to	(4)	2
		(5)	3

Code:

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 5$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 1$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 2, 4; R \rightarrow 4; S \rightarrow 5$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

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-
1. A child has n coins of Rs. 1. Each day he buys exactly one of the three objects either pencil for Rs. 1 or pen for Rs. 2 or book for Rs. 2. If S_n denotes the number of ways of spending n rupees, then the value of $\left(\frac{S_{11} - S_{10}}{S_9} + S_7 \right)$ is _____.
 2. Let incircle of radius 4 units of a triangle ABC touches the side BC at D. If $BD = 6$, $DC = 8$ and Δ be the area of triangle, and R be radius of circumcircle of ΔABC , then $\sqrt{\Delta - 3} + R$ is equal to _____.
 3. On a semicircle with diameter AB and centre O point E and C are marked in such a way that OE is perpendicular to AB, and the chord AC intersects the segment OE at a point D which is interior to the semicircle. If $\angle CBA = \frac{\pi}{k}$ such that a circle can be inscribed into the quadrilateral OBCD then value of k is equal to _____.
 4. If $z_1, z_2, z_3 \in \mathbb{C}$ satisfy the system of equations given by $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ and $z_1 z_2 z_3 = 1$ such that $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$ then find the value of $|z_1 + z_2^2 + z_3^3|$ is _____.

5. If $\sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\cot^{-1}(n^2 + 3n + 3)}{1 + \cot^{-1}(n+1) \cot^{-1}(n+2)} \right) = \tan^{-1} \left(\frac{p}{4} \right)$ then find the value of $\cos^{-1}(\cos(p-1))$
[Use : $\pi = 3.14$]
6. Given that $f(x)$ is a function defined on $\mathbb{R}$, satisfying $f(1) = 1$ and for any $x \in \mathbb{R}$, $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = f(x) + 7 - x$, then $g(2023)$ is equal to _____
7. Let $f(x) = 2x(2-x)$, $0 \leq x \leq 2$. If the number of roots and sum of all roots of equation $f(f(f(x))) = \frac{x}{2}$ are equal to 'a' and 'b' respectively, then $(b+a)$ is equal to _____
8. Tangent is drawn at the point (x_i, y_i) on the curve $y = f(x)$, which intersects the x -axis at $(x_{i+1}, 0)$. Now again a tangent is drawn at (x_{i+1}, y_{i+1}) on the curve which intersects the x -axis at $(x_{i+2}, 0)$ and the process is repeated 'n' times. If $x_1, x_2, x_3, \dots, x_n$ form a G.P. with common ratio 'r' and curve $y = f(x)$ is a hyperbola with eccentricity equal to 'e' then $r + e^2$ is equal to _____

SPACE FOR ROUGH WORK

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

SPACE FOR ROUGH WORK

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